Applying PMI Project Management Framework to Software Validation of LC/MS Systems

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Introduction

The Project Management Framework developed by the Project Management Institute (PMI) is recognized as the gold standard for good practice in project management especially for large projects.[1] But the Framework is also valuable for managing small projects.

This paper describes how the Framework can be applied to Software Validation of LC/MS Systems. LC/MS stands for Liquid Chromatography-Mass Spectrometry. LC/MS systems are used for drug discovery, forensic analysis, food quality analysis and other safety-critical research. This is a good example because software validation of LC/MS systems occurs in the GLP laboratory. Project management is critical because GLP laboratories must comply with strict Federal Drug Administration (FDA) regulations. Key challenges include risk management, project planning and quality management. These challenges are common for anyone conducting a small project especially those required to meet government regulations.

The specific example in this paper is software validation of the Analyst® software designed for controlling the mass spectrometers (MS) produced by AB SCIEX. These projects are usually two or three weeks in duration but have a critical impact because patient and environmental safety often depends on accurate lab results.

The AB SCIEX team chose the PMI Project Management Framework because it provides excellent guidance for mitigating the important challenges of risk management, project planning and quality management. Their usage of the Framework has been so successful that each member of the team found it advantageous to attain a PMP® certificate. After one software validation project, an FDA auditor used "utmost professionalism" and "impeccable organization" to describe the results. Clearly, the PMI Project Management Framework is a valuable tool for these and other small projects.

The following sections describe in detail how the Framework is used. First, GLP is defined. Then, an overview of software validation in accordance with GLP and GAMP® 5 is provided. Next, there is an synopsis of the PMI Project Management Framework. Then, the challenges for software validation for LC/MS Systems are discussed in the context of the PMI Framework.

Overview

GLP

GLP stands for "Good Laboratory Practice" as defined by Organization for Economic Co-operation and Development (OECD)[2]. Good Laboratory Practice (GLP) embodies a set of principles that provide a framework within which laboratory studies are planned, performed, monitored, recorded, reported and archived. These studies are generally undertaken for pharmaceuticals (only pre-clinical studies), agrochemicals, cosmetics, food additives, detergents, etc to generate data for assessing the hazards to consumers and third parties, including the environment. GLP helps assure regulatory authorities that the data submitted are a true reflection of the results obtained during the study and can therefore be relied upon when making risk/safety assessments.

GAMP®

GAMP® 5 is the latest version of Good Automated Manufacturing Practice developed by the International Society for Pharmaceutical Engineering (ISPE) [3]. ISPE has also issued guidelines specific to the pharmaceutical industry called *The Good Automated Manufacturing Practice (GAMP®) Guide for Validation of Automated Systems in Pharmaceutical Manufacture.* This guide describes a set of principles and procedures that help ensure that pharmaceutical products have the required quality.

One of the core principles of GAMP® is that quality must be built into each stage of the project. It cannot wait until the end.

The most important update in GAMP® 5 is the focus on risk management as it relates to patient safety.[4] GAMP® 5 requires validation if there

could be an impact on patient safety, product quality, or data integrity.[5] Therefore, the decision to validate, what to validate and how to validate is largely an exercise in risk management.

Software Validation

Rules and regulations for software validation of laboratory systems can be found in 21 CFR 211 (Good Manufacturing Practice), 21 CFR 58 (Good Laboratory Practice), 21 CFR 820 (Quality System Regulation for Medical Devices), 21 CFR Part 11 (Electronic Records and Electronic Signatures), as well as several others. [6,7] An analysis of the relevant regulations is beyond the scope of this paper. However, the spirit of the regulations can be summarized in the FDA definition:

> *"confirmation by examination and provision of objective evidence that software specifications conform to user needs and intended uses, and that the particular requirements implemented through software can be consistently fulfilled." [4]*

Confirmation of conformity to user needs and intended uses is obtained by comparing actual system performance against pre-determined requirements. During validation this comparison is accomplished by executing test procedures and collecting objective evidence (screen shots, printed reports, result data files, etc.). The goal is to demonstrate that the software satisfies the intended use. This can only be accomplished if validation activities were properly planned and the tests were executed according to the plan. Therefore, a project management framework is critical to the success of software validation of LC/MS Systems.

Synopsis
PMI Project Management Framework [1]

The PMI Project Management Framework is described in the PMBOK® Guide (*A Guide to the Project Management Body of Knowledge*). According to the PMBOK® Guide, project management is the application of knowledge, skills, tools and techniques to project activities to meet the project requirements. This framework includes the Process Groups and Knowledge Areas defined below.

Process Groups

The framework includes the five Process Groups:

1. **Initiating Process Group:** Processes required to define a new project
2. **Planning Process Group:** Processes required to establish the scope of the project and define the course of action required to attain the project objectives
3. **Executing Process Group.** Process performed to complete the work defined in the project management plan to satisfy the project objectives
4. **Monitoring and Controlling Process Group.** Process required to track and review the progress and performance of the project
5. **Closing Process Group.** Processes performed to finalize all activities to formally close the project

Knowledge Areas

The framework also includes the following Knowledge Areas. *(Note, in the PMBOK®, these are numbered beginning with 4 to correspond with the section of the guide where the knowledge area is discussed.)*

4. Project Integration Management – includes the processes and activities needed to identify, define and coordinate various project management activities.

5. Project Scope Management – processes required to ensure that the project includes the work required, and only the work required, to complete the project.

6. Project Time Management – processes required to manage timely completion of the project.

7. Project Cost Management – processes involved in estimating, budgeting and controlling costs.

8. Project Quality Management – processes and activities to ensure quality control.

9. Project Human Resource Management – processes for organizing, managing, and leading the project team.

10. Project Communications Management – processes required to ensure timely and appropriate generation, collection, distribution, storage, retrieval, and ultimate disposition of project information.

11. Project Risk Management – processes for conducting risk planning, identification, analysis, response planning and monitoring and control.

12. Project Procurement Management – processes necessary to purchase or acquire products, services or results needed from outside the project team.

Additionally, there are two terms used frequently by PMBOK® that require special definitions:

- Organizational Process Assets – templates, policies, procedures, guidelines, etc, of an organization
- Enterprise Environmental Factors – government regulations, industry standards, economic conditions…

The PMI Project Management Framework is an excellent framework for software validation of LC/MS Systems because it is a highly regarded, well-crafted approach that contains many useful tools and techniques for successful project management. Also, use of this framework produces the artifacts necessary to prove compliance with GAMP® 5, 21 CFR Part 11 (Electronic records and electronic signatures) and other regulations.

Challenges For Software Validation of LC/MS Systems

As discussed above risk management is a key challenge for project management of software validation of LC/MS Systems in accordance with GAMP® 5. Other important project management challenges include: planning the software validation project and assuring quality.

The Project Management Framework provides guidance for these challenges. This guidance is addressed in the following sections:

- Project Risk Management
- Project Planning
- Quality Management

Project Risk Management

PMBOK® Chapter 11 titled "Project Risk Management" defines risk as *an uncertain event that if it occurred would effect a project objective such as scope, schedule, cost and/or quality*. Quality is most important in the GLP laboratory because of the safety-critical nature of work being conducted. Software validation of LC/MS Systems must address the risk to patient safety described in GAMP® 5. In the GLP laboratory, possible hazards to patients can result from errors in tests such as drug testing or forensic analysis or from errors in experiments such as discovery of new medications, development of food additives, etc. Therefore, conducting and documenting the software Validation Risk Assessment (VRA) is critical.

PMBOK® Chapter 11 contains guidance in six Knowledge Areas that can be used to assess and to develop responses for the risk to patient safety addressed by GAMP® 5. The first five Knowledge Areas fall under the Planning Process Group. The sixth Knowledge Area is in the Monitoring and Controlling Process Group. The Risk Management Knowledge Areas are described below:

11.1 Plan Risk Management – process of defining how to conduct risk management activities

11.2 Identify Risks – process of determining risks and documenting their characteristics

11.3 Perform Qualitative Risk Analysis – process of prioritizing risk for further analysis or action

11.4 Perform Quantitative Risk Analysis – process of numerically analyzing the effect of identified risks on the overall project objectives

11.5 Plan Risk Responses – process of developing risk mitigations

11.6 Monitor and Control Risks – process of implementing risk response plans, tracking identified risks, identifying new risks and evaluating risk process effectiveness throughout the project

Each of these processes is discussed below as it relates to software validation for LC/MS Systems.

PMBOK® 11.1 Plan Risk Management

A key component of the Risk Management Plan is the definition of the probability and impact matrix. GAMP® 5 prescribes guidelines for completing the probability and impact matrix that follow the spirit of the PMBOK® guidance but are slightly different. In GAMP® 5 prioritizing risk is a two step process with two matrices from the GAMP® 5, A Risk-Based Approach to Compliant GxP Computerized Systems. The first matrix is probability and severity and the second is detect-ability. These two matrices are described below.

To begin Probability is determined. Probability is the chance of an event occurring. Rather than the actual percentages prescribed by PMBOK®, there are three classifications for Probability:

Low – The chance of the event occurring is almost impossible

Medium – The probability of the event occurring is possible but not likely

High – The event will most likely occur

Next, the Severity is defined. Severity is the effect of an adverse event in terms of short and long term impact on the business and regulatory compliance. Again rather than actual percentages, there are three classifications of Severity:

Low – event is expected to have a minor negative impact. The detrimental effects would not be long-term.

Medium – the event is anticipated to have a moderate impact with short- to medium-term detrimental effects

High – the event will damage the SYSTEM with long-term effects

Once the Probability and Severity have been decided then the risk can be classified based on the matrix in Figure 1.

Next it is necessary to define Detect-ability. Detect-ability is the likelihood that an adverse event can be detected. There are three classifications for detect-ability:

Low – Detection of the fault condition is perceived to be unlikely

Medium – Detection of the fault condition is perceived to be reasonably likely

High – Detection of the fault condition is perceived to be highly likely

By combining the Risk Class with the Detect-ability, a priority level can be identified for each risk as shown in Figure 2.

Figure 1: Probability Severity Matrix

Severity \ Probability	Low	Medium	High
High	Risk Class 2	Risk Class 1	Risk Class 1
Medium	Risk Class 3	Risk Class 2	Risk Class 1
Low	Risk Class 3	Risk Class 3	Risk Class 2

Severity - impact on patient safety, product quality and data integrity (or other harm)

Probability - likelihood of the fault occurring

Risk Class - Severity versus Probability

Figure 2: Detect-ability and Risk Class Matrix

Risk Classes \ Detect-ability	High	Medium	Low
1	Medium Risk Priority	High Risk Priority	High Risk Priority
2	Low Risk Priority	Medium Risk Priority	High Risk Priority
3	Low Risk Priority	Low Risk Priority	Medium Risk Priority

Detect-ability - likelihood that the fault will be noted before harm occurs

Risk Priority - Risk Class versus Detect-ability

PMBOK® 11.2 Identify Risks

The Risk Register contains a list of identified risks. A sample Risk Register for software validation is shown in Appendix A. It was developed using document reviews, brainstorming, interviewing, checklist analysis, assumptions analysis, SWOT and expert judgment explained in the PMBOK®.

PMBOK® 11.3 Perform Qualitative Risk Analysis

Perform Qualitative Risk Analysis is the process of following instructions in risk management plan for prioritizing risks. At this point the Project Management Framework and GAMP® 5 converge. According to each, once the priority level has been assessed then validation efforts can focus on mitigating risks of high criticality first.

PMBOK® 11.4 Perform Quantitative Risk Analysis

Perform Quantitative Risk Analysis is the process of numerically analyzing the effect of identified risks on the overall project objectives. It is performed on risks which have been identified as likely to occur and have a substantial impact.

PMBOK® 11.5 Plan Risk Responses

During Plan Risk Responses expert judgment is used to devise methods to avoid threats and exploit opportunities. These are documented in the software validation risk register in Appendix A.

PMBOK® 11.6 Monitor and Control Risks

Risk re-assessment is conducted throughout the software validation project because new risk can become apparent as the validation engagement progresses. An example of a risk that occurs during software validation of Analyst® is failure to acquire sufficient samples for the load test. Failure can be the result of tiny deviations in peripheral machine settings that magnify over the duration of the test. For example, the position of the needle over the vial can change until the needle hits the edge of the vial and stops the load test.

Project Planning

Regulatory auditors for GLP laboratories look closely at the Validation Plan and the Test Plan to ensure completeness. Guidance for project planning can be found in the Planning Process Group. The following project management Knowledge Areas fall into this group:

4.2	Develop Project Management Plan
5.1	Collect Requirements
5.2	Define Work Breakdown Structure
6.1	Define Activities
6.2	Sequence Activities
6.3	Estimate Activity Resources
6.4	Estimate Activity Durations
6.5	Develop Schedule
7.1	Estimate Costs
7.2	Determine Budget
8.1	Plan Quality (Quality Management section)
9.1	Develop Human Resource Plan
10.1	Plan Communications
11.1	Plan Risk Management (above)
11.2	Identify Risks (above)
11.3	Perform Qualitative Risk Analysis (above)
11.4	Perform Quantitative Risk Analysis (above)
11.5	Plan Risk Responses (discussed above)
12.1	Plan Procurements

Each of these processes is discussed below as it relates to software validation of LC/MS Systems.

PMBOK® 4.2 Develop Project Management Plan

The Project Management Plan is developed using the project charter, results of other planning processes, enterprise environmental factors, and organizational process assets. For software validation of Analyst® the contractual Statement of Work (SOW), rather than a project charter, includes the scope, time, cost, human resources, communication and procurement information.

Enterprise environmental factors for software validation for LC/MS Systems include regulations such as 21 CFR Part 11 (Electronic records and electronic signatures) and FDA regulations (or equivalent), as well as, industry standards like GAMP® 5 and GLP.

Organizational process assets include document templates of the organization.

PMBOK® 5.1 Collect Requirements

The Collect Requirements process results in a list of requirements, a requirements management plan, and a traceability matrix.

GAMP® 5 aligns with PMBOK® 5.1 Collect Requirements in that both discuss collecting user and functional requirements. For software validation purposes these requirements can be combined into the User Functional Requirements Specification (UFRS). The Requirements Management Plan is also documented in the UFRS.

The UFRS also states the criteria that will be used to judge the overall suitability of the system. The UFRS must address Technical Controls, Procedural Controls, capacities, accuracy, security, fault tolerance, physical environment, and training requirements, among others. It is critical that the UFRS be a complete statement of the needs and objectives of the acquiring organization. A typical UFRS for software validation of LC/MS Systems will contain up to several hundred unique requirements.

The definition of the Requirements Traceability Matrix is also the same for PMBOK® and GAMP® 5. This matrix shows the relationship between each user/functional requirement and the corresponding test script (in the case of technical controls) or Standard Operating Procedure (SOP) (for procedural controls). The traceability matrix makes it possible to confirm that each user requirement has been addressed and satisfied by the validation procedure.

PMBOK® 5.2 Define Scope

For software validation for LC/MS Systems, expert judgement is generally used to define scope. Scope is documented in the SOW and included as part of the contract.

PMBOK® 5.3 Create WBS (Work Breakdown Structure)

For software validation of LC/MS Systems expert judgment can be used to decompose the following task list into work packages.

1. Develop Project Management Plan
2. Develop Document Approval Process
3. Conduct Validation Risk Assessment
4. Conduct 21 CFR Part 11 Assessment
5. Identify and document the User Functional Requirements Specification (UFRS)
6. Design and document the System Design Specification (SDS)
7. Develop Traceability Matrix (TM)
8. Develop Test Plan (TP)
9. Develop Installation Qualification (IQ)
10. Develop Operational Qualification (OQ)
11. Develop Performance Qualification (PQ)
12. Conduct Installation Qualification (IQ)
13. Conduct Operational Qualification (OQ)
14. Conduct Performance Qualification (PQ)
15. Update SOP
16. Write Validation Summary Report (VSR)

PMBOK® 6.1 Define Activities

Define Activities results in an activity list and a milestone list. Further decomposition of WBS work packages yields the following activity list for software validation of LC/MS Systems. A milestone list follows the activity list.

Activity List

1. Develop Project Management Plan
    a. Review SOW
    b. Review summary milestone list
    c. Review summary budget
    d. Review governmental regulations (FDA...)
    e. Review lessons learned on past projects
    f. Identify location of laboratory equipment
    g. Identify company guidelines
    h. Review performance measurement criteria
    i. Identify PMIS or project management templates
    j. Identify change control procedures
    k. Identify configuration management systems
2. Conduct Validation Risk Assessment
    a. Plan risk management
    b. Identify risks
    c. Perform Qualitative Risk Analysis
    d. Perform Quantitative Risk Analysis
    e. Plan Risk Responses
3. Conduct 21 CFR Part 11 Assessment
    a. Review 21 CFR Part 11
    b. Plan electronic records and electronic signatures management
    c. Identify electronic records
    d. Identify which document require electronic signatures

4. Identify and document the User Functional Requirements Specification (UFRS)
    a. List user requirements
    b. List functional requirements
5. Design and document the System Design Specification (SDS)
    a. Identify laboratory equipment
    b. Identify computer hardware
    c. Identify computer software
    d. Design the physical system
    e. Design the data flow within the system
    f. Design the system configuration
6. Develop Traceability Matrix (TM)
    a. Match requirements to test cases
7. Develop Test Plan (TP)
    a. Match test cases to requirements
8. Develop Installation Qualification (IQ)
    a. Develop test scripts to meet each installation requirement
9. Develop Operational Qualification (OQ)
    a. Develop test scripts to meet each operational requirement
10. Develop Performance Qualification (PQ)
    a. Develop test scripts to meet each performance requirement
11. Conduct Installation Qualification (IQ)
    a. Execute test scripts
    b. Document results
    c. Identify anomalies, if any
    d. Identify deviations, if any
12. Conduct Operational Qualification (OQ)
    a. Execute test scripts
    b. Document results
    c. Identify anomalies, if any
    d. Identify deviations, if any
13. Conduct Performance Qualification (PQ)
    a. Execute test scripts
    b. Document results
    c. Identify anomalies, if any
    d. Identify deviations, if any
14. Update Standard Operating Procedures
    a. Identify SOPs
    b. Identify required updates
    c. Document updates
15. Write Validation Summary Report (VSR)
    a. Document results of validation
    b. Document anomalies, if any
    c. Document deviations, if any

For Analyst® lower level activities are detailed in the test plan and validation protocols (IQ, OQ, PQ). Each validation protocol contains test scripts. Each test script contains step-by-step instructions for executing a test including collecting the objective evidence to show that the governing requirement was met.

Milestone List

- Delivery of Vendor Assessment (or completion of vendor audit)
- Delivery of Validation Plan (VP)
- Delivery of Validation Risk Assessment (VRA)
- Delivery of User Functional Requirements Specification (UFRS)
- Delivery of System Design Specification (SDS)
- Delivery of Test Plan
- Delivery of Installation Qualification Protocol
- Delivery of Operational Qualification Protocol
- Delivery of Performance Qualification Protocol
- Delivery of 21 CFR Part 11 Assessment
- Delivery of Traceability Matrix
- QA Approval of software Validation Document set
- Completion of Installation Qualification Protocol
- Completion of Operational Qualification Protocol
- Completion of Performance Qualification Protocol
- QA Approval of completed protocols
- Completion of QAU Review
- Delivery of Validation Summary Report (VSR)

PMBOK® 6.2 Sequence Activities

For software validation of GLP laboratories sequencing of activities is done via Precedence Diagramming Method (PDM). PDM is used to construct a project schedule network diagram. Microsoft Project is adequate for documenting

the sequences of activities for software validation of LC/MS Systems.

Dependency determination is based on expert judgment. Dependencies are generally "Finish-to-Start". Mandatory dependencies are denoted in the contract or inherent in the nature of the work. For example, audit trail review must occur after an audit trail event happens. External dependencies generally occur because the dependency on IT network technicians who are typically responsible for multiple projects.

Protocols are sequenced per the GAMP® 5 lifecycle so that the Installation Qualification (IQ) protocol must be completed and approved before the Operational Qualification (OQ) protocol can be executed. And the OQ must be completed and approved before the Performance Qualification (PQ) can be done. Additionally, within each of the protocols, test scripts have numbered steps that must be done in order.

For sake of brevity, the activity list in the section above was sequenced.

PMBOK® 6.3 Estimate Activity Resources

Workers with special skills are required to conduct software validation of LC/MS Systems and the equipment to be validated must be available. The resource breakdown structure can include the following:

- Tester with knowledge of Analyst® software, AB SCIEX mass spectrometers, Microsoft Windows, Microsoft Office and computer networks
- Mass spectrometer operator trained in scientific studies, GLP and AB SCIEX mass spectrometers
- IT personnel trained in computer network technology and Analyst Administrator Console (AAC), if used.
- QA personnel trained in quality control and knowledgeable about scientific studies in the GLP environment
- Mass spectrometer and peripheral equipment must be scheduled for software validation

Expert judgment and bottom-up estimating are generally used to estimate activity resources.

PMBOK® 6.4 Estimate Activity Durations

For software validation of GLP laboratories, activity durations are generally estimated based on expert judgment and analogous estimating of prior, similar projects. Three-point estimates based on PERT have also been used.

PMBOK® 6.5 Develop Schedule

For software validation of LC/MS Systems, the schedule is developed using the critical chain method because resources are limited due to the special skill level required. Resource levelling is used especially with network technicians who generally work on other projects during the software validation.

Lags may be applied because of the uncertainty of quality review resulting because quality reviewers may not be available exactly when each phase of the project is complete.

Additionally, schedule crashing is often used to accelerate the software validation project so that "validated" equipment can be used for GLP studies more quickly.

PMBOK® 7.1 Estimate Costs

For software validation of LC/MS Systems costs are estimated based on prior projects, vendor bid analysis, marketplace needs, bottom-up estimating and the cost of quality.

PMBOK® 7.2 Determine Budget

For software validation of LC/MS Systems budgets are estimated using expert judgment, lessons learned from prior projects and the needs of the marketplace. Cost aggregation is sometimes used for customized software validation projects.

PMBOK® 9.1 Develop Human Resource Plan

Workers with special skills are required to conduct software validation of LC/MS Systems. Therefore, resource availability is a critical factor in developing a human resource plan. The project organization chart is matrix-based much like RAM. Formal and informal networking is important. Because GLP laboratories are located around the world, it is important to understand the culture of the organization and any additional governmental regulations.

PMBOK® 10.2 Plan Communications

Software validation of LC/MS Systems are frequently conducted by virtual teams who prepare "offsite" then key team members meet at the laboratory to conduct the validation. Therefore, email is an important communications tool. It is used to communicate routine status, to ask questions, to route documents for review, and to record results of meetings. "Texting" is also becoming a good way to communicate status or to escalate concerns. Thus the Blackberry, iPhone or similar devices are quickly becoming a vital tools.

Telecons may be used to initiate the project but workers in different geographic locations and in different time zones make it difficult to schedule team telecons on a routine basis.

Typically status is reported to the core project team on a daily basis. Management receives news when milestones are completed or if expert resources are needed to answer questions or to address unexpected events.

PMBOK® 12.1 Plan Procurements

The software validation procurement decision is essentially a "make-or-buy" decision. Each company must decide whether to use in-house resources or the hire a third party to validate the software. Expert judgment is key to making this decision.

When selecting a third party it is important to remember that GAMP® 5 recognizes that most computerized systems are now based on configurable packages that utilize computer networks. Therefore, Supplier Audit Programs have more importance in GAMP® 5 because more and more supplier certificates are accepted in lieu of actual supplier audits. AB SCIEX provides such a certificate called a Software Structural Validation Certificate. It states that the Analyst® product was developed and structurally validated following a Registered Quality Management System which conforms to ISO 9001 International Standard requirements.

A firm fixed price contract is generally used for Analyst® software validation because AB SCIEX can rely upon past experience to properly estimate the cost of the validation effort. However, if the software validation project is highly customized, a T&M contract may be used.

Quality Management

Quality is a primary concern of software validation of LC/MS Systems because the studies conducted at these laboratories can affect human safety. Failure to achieve quality goals could result in harm or even death.

The Project Management Framework defines quality as *"the degree to which a set of inherent characteristics fulfil requirements"*. For software validation of LC/MS Systems, the criteria for quality is inherent in the term "GLP laboratory". By definition a GLP laboratory must meet FDA regulations. Thus quality is defined as meeting FDA regulations.

The following three components of quality management are described in the PMBOK® Knowledge Area 8.0 Quality Management:

- PMBOK® 8.1 Plan Quality
- PMBOK® 8.2 Perform Quality Assurance
- PMBOK® 8.3 Perform Quality Control

Note: Quality management described in the Project Management Framework is intended to be compatible with the International Organization for Standardization (ISO).

PMBOK® 8.1 Plan Quality

For software validation of LC/MS Systems, cost benefit analysis and expert judgment are the primary tools for planning quality management. Statistical sampling is not used because 100% of the requirements must be tested.

PMBOK® 8.2 Perform Quality Assurance

During the software validation of LC/MS Systems, the Quality Assurance (QA) department must be actively engaged in the validation effort. Management's approval of validation depends on the recommendation of QA.

Each software validation document, protocol and test result must be signed by QA. To facilitate this, the software validation package includes a Quality Assurance Review document to ensure completeness of the quality review. It lists each deliverable, the status (either pass or fail), official document name, reviewers and approvers, approval dates and any comments. This must be completed before the software can be deemed "validated".

Fortunately, GAMP® 5 simplified the document approval process. Technical experts are now empowered to approve technical documentation while QA must ensure documents meet applicable regulations. For example QA should review a UFRS against the applicable regulations but the UFRS technical review is the responsibility of technical subject matter experts. QA should verify that design specifications are being produced for projects (i.e. verify that processes are being followed) but QA does not need to sign every technical document in a project.[1]

PMBOK® 8.3 Perform Quality Control

Inspection is the tool most frequently used in software validation of LC/MS Systems. Approved change requests are also reviewed by QA.

Conclusion

The Project Management Institute (PMI) Project Management Framework provides excellent guidance for software validation of LC/MS Systems. The example clearly demonstrates how the Framework is used to mitigate key challenges of risk management, project planning and quality management describing specific guidance from the Process Groups and Knowledge Areas addressing these concerns.

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Biographies



Ms. Stacy D. Nelson, PMP® is a project manager who brings her prior software validation experience at NASA to the GLP laboratories around the world. She has published over 20 technical papers about software verification and validation for safety-critical software, as well as, co-authoring a book titled, *Independent Verification and Validation of Neural Networks*. She recently won the DNA Award for **D**emonstrated **N**oteworthy **A**chievement for the AB SCIEX core value, "The Best Team Wins".

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As the Senior Manager for the AB SCIEX Compliance Services and Validation Consulting organization, Dave has led the group since its inception in 2006. He currently travels the globe to work with AB SCIEX's mass spectrometry users in mapping out and designing compliance strategies. For nine years prior, Dave was with Cisco Systems in various IT and Operations roles, where he achieved his PMP® certification. Dave also has extensive experience in the financial derivatives markets working for various banks and trading firms in New York and Chicago. Dave has a BA in Economics and an MBA from the University of Texas at Austin.

Appendix A: Sample Risk Register

Function	Sub-function	Relevance (GxP/Bus.)	Risk Scenario	Probability	Severity	Class	Detect-ability	Priority	Mitigation
Security	Access	Both	Unauthorized access to or modification of data	Medium	High	1	Low	High	SOP will specify access controls and password policy.
	Access	Both	Unauthorized privileges	Medium	Medium	2	Medium	Medium	SOP will instruct administrator to confirm all privileges assigned to users.
	Data	GxP	Deletion of data before generation of raw data	Low	High	2	High	Low	System Operation SOP specifies the correct operating procedures; training will be provided to operators to prevent incorrect operation.

SOP – Standard Operating Procedure

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